Python ala 2

1. What are Data Types in Python?

Definition:

Data types define the **type of value a variable can store** and the **operations allowed on it**.

Explanation:

Python automatically assigns data types at runtime (dynamic typing). This helps manage memory and ensures correct operations.

Syntax:

```
x = 10
y = "Hello"
```

Example:

```
x = 5
print(type(x))   # <class 'int'>
```

2. What is Dynamic Typing?

Definition:

Dynamic typing means **no need to declare data type explicitly**.

Explanation:

Python decides the type based on the assigned value during execution.

Syntax:

```
x = 10
x = "Python"
```

Example:

```
x = 100
x = "Now I am string"
print(type(x))
```

3. What is Strong Typing?

Definition:

Python does not allow **implicit conversion between incompatible types**.

Explanation:

You must convert types manually to avoid errors.

Syntax:

```
int("5")
```

Example:

```
# print("5" + 5) → Error
print(int("5") + 5)  # 10
```

4. What are Numeric Data Types?

Definition:

Used to store numeric values like integers, decimals, and complex numbers.

Explanation:

Includes:

- int
- float
- complex

Syntax:

```
a = 10
b = 2.5
c = 3 + 4j
```

Example:

```
print(type(3.14))  # float
```

5. What is a String?

Definition:

A sequence of characters enclosed in quotes.

Explanation:

Strings are immutable and support many operations like slicing and formatting.

Syntax:

```
name = "Python"
```

Example:

```
print(name.upper())  # PYTHON
```

6. What is a List?

Definition:

An ordered and mutable collection of elements.

Explanation:

Lists can store different data types and allow modification.

Syntax:

```
my_list = [1, 2, 3]
```

Example:

```
my_list.append(4)
print(my_list)
```

7. What is a Tuple?

Definition:

An ordered but immutable collection.

Explanation:

Once created, values cannot be changed.

Syntax:

```
my_tuple = (1, 2, 3)
```

Example:

```
print(my_tuple[0])
```

8. What is a Set?

Definition:

An unordered collection of unique elements.

Explanation:

Duplicates are automatically removed.

Syntax:

```
my_set = {1, 2, 3}
```

Example:

```
print({1,2,2,3})  # {1,2,3}
```

9. What is a Dictionary?

Definition:

A collection of key-value pairs.

Explanation:

Used for fast data retrieval using keys.

Syntax:

```
student = {"name": "Harshil", "age": 20}
```

Example:

```
print(student["name"])
```

10. What is Type Conversion?

Definition:

Process of converting one data type into another.

Explanation:

Two types:

- Implicit (automatic)
- Explicit (manual)

Syntax:

int(), float(), str()

Example:

```
a = "10"
b = int(a)
print(b + 5)  # 15
```